

LiDAR Surface Vulnerability Testing Analysis Report

Dissertation context: "Hacking Non-GPS/GPS-denied Drones: Security Analysis of LiDAR-based UAV Navigation"

Sensor: Garmin LiDAR-Lite v4 (1D) Platform: Holybro X500 UAV Autopilot: PX4

Generated from the uploaded experimental result images and supporting planning documents.

Executive Summary

This report analyses four LiDAR surface-manipulation test sets relevant to GPS-denied UAV navigation using a Garmin LiDAR-Lite v4 mounted for PX4-based use. The objective was to determine whether transparent, reflective, mirror-like, and optically altered surfaces can induce measurement bias, false ranging, or outright signal loss that could be security-relevant for altitude estimation, landing, and obstacle avoidance. The empirical results show three distinct failure classes. First, the transparent-surface condition produced a **stable but biased** overestimate: the mean reading was 36.63 cm against an actual distance of 30.1 cm, equivalent to a mean signed error of 6.53 cm (21.71%). Second, the reflective/aluminium-foil set showed **angle-dependent degradation**: mean readings increased from 43.90 cm at 0° to 86.87 cm at 60°. Third, the mirror-based conditions demonstrated **catastrophic false ranging** and, under magnification, **signal loss**. The single worst measured regime was Test 3 at 15°, with RMSE 303.80 cm; the highest mean bias was 303.79 cm. In Test 4, the 15°-45° range was annotated as signal-quality loss, implying an effective measurement failure rate of 100% for those angles. For dissertation purposes, the key security interpretation is that single-beam LiDAR can be manipulated not only into modest inaccuracy, but into qualitatively different error modes: believable bias, gross over-ranging, and no usable return. These modes can each disrupt PX4 logic if fused without appropriate confidence management, temporal validation, and cross-sensor checks.

Introduction

LiDAR-based navigation is attractive in GPS-denied or GPS-degraded UAV missions because it offers direct geometric information for altitude holding, terrain following, proximity awareness, and landing support. In platforms such as the Holybro X500 running PX4, a rangefinder can become particularly influential during indoor flight, under structures, or in other contexts where GNSS is unreliable or unavailable. However, optical ranging is not purely a software problem: it is mediated by material properties, incidence angle, reflectivity, and ambient optical conditions. This makes the sensor not only an environmental perception component, but also a potential attack surface. From a security perspective, a malicious or simply adverse environment may induce a false perception of free space, a false nearby obstacle, or a spurious altitude estimate. A stable bias can be dangerous because it looks plausible; catastrophic false ranging is dangerous because it can inject large state errors; and signal loss is dangerous because downstream logic must decide whether to extrapolate, reject, or fail safe. The dissertation theme therefore sits at the intersection of cyber-physical security and perception robustness: how material and optical manipulation can compromise autonomous behaviour when the system must rely on LiDAR in lieu of GPS.

Methodology

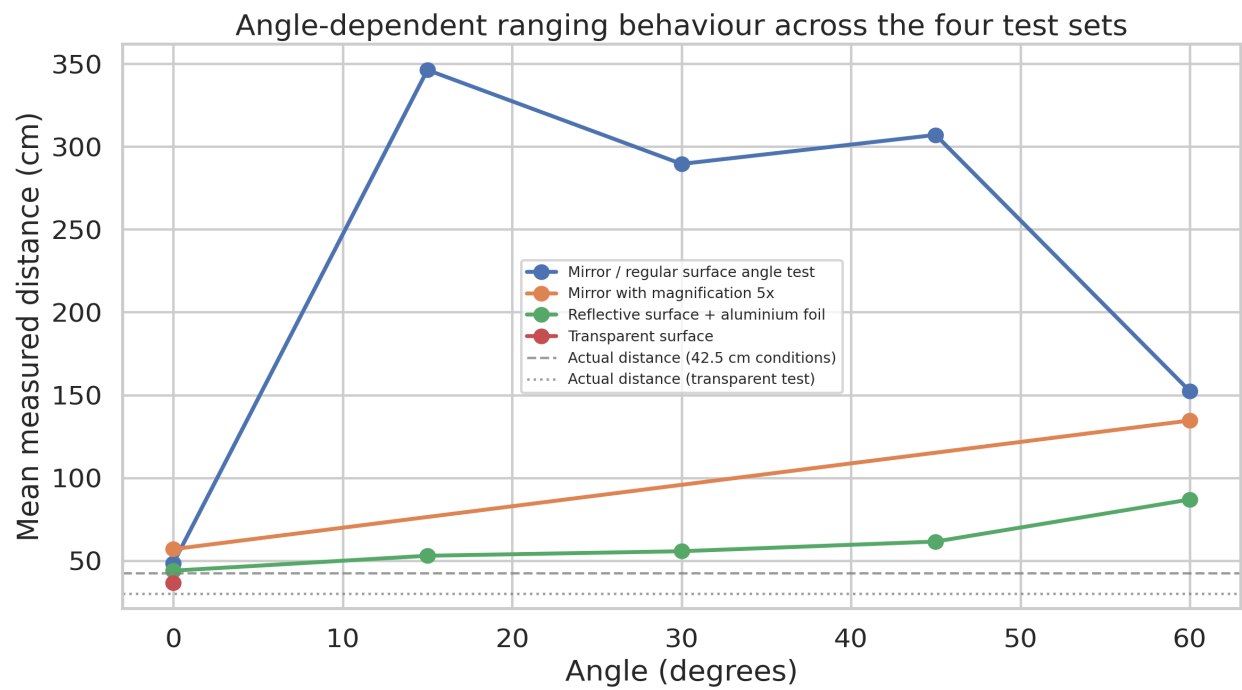
The analysis is based on four uploaded result tables captured as images. The sensor under study is the Garmin LiDAR-Lite v4, a 1D optical rangefinder commonly integrated into UAV systems. The experimental context provided by the user identifies a Holybro X500 UAV platform and PX4 autopilot. Each test set consisted of repeated scalar range measurements under controlled bench conditions with a known ground-truth separation between sensor and target condition. The surface conditions varied across transparent, reflective, mirror-like, and magnified optical arrangements. Test Set 1 examined a reflective-surface/aluminium-foil configuration at an actual distance of 42.5 cm across 0°, 15°, 30°, 45°, and 60° incidence angles. Test Set 2 examined a transparent surface at 0° with an actual distance of 30.1 ± 0.1 cm. Test Set 3 examined a mirror/regular-surface angular configuration at 42.5 cm across the same angle set. Test Set 4 examined a mirror with 5× magnification at 42.5 cm; the uploaded notes state that signal quality is lost from 15° to 55°, while readings at 60° and above tend toward a higher apparent distance. For each condition, the analysis computed mean, standard deviation, variance, minimum, maximum, mean signed error, mean absolute error, mean percentage error, mean absolute percentage error, RMSE, and failure rate. Where the image explicitly stated signal loss rather than numeric returns, the condition was retained as a failed regime rather than silently excluded.

Results Overview

Test set	Angle	Actual (cm)	n valid	Failure %	Mean	Std	Min	Max	Bias	MAE	MAPE %	RMSE	Classification
Test 1	0°	42.50	31	0.00	43.90	0.70	43.00	45.00	1.40	1.40	3.30	1.56	Near-normal return with mild bias

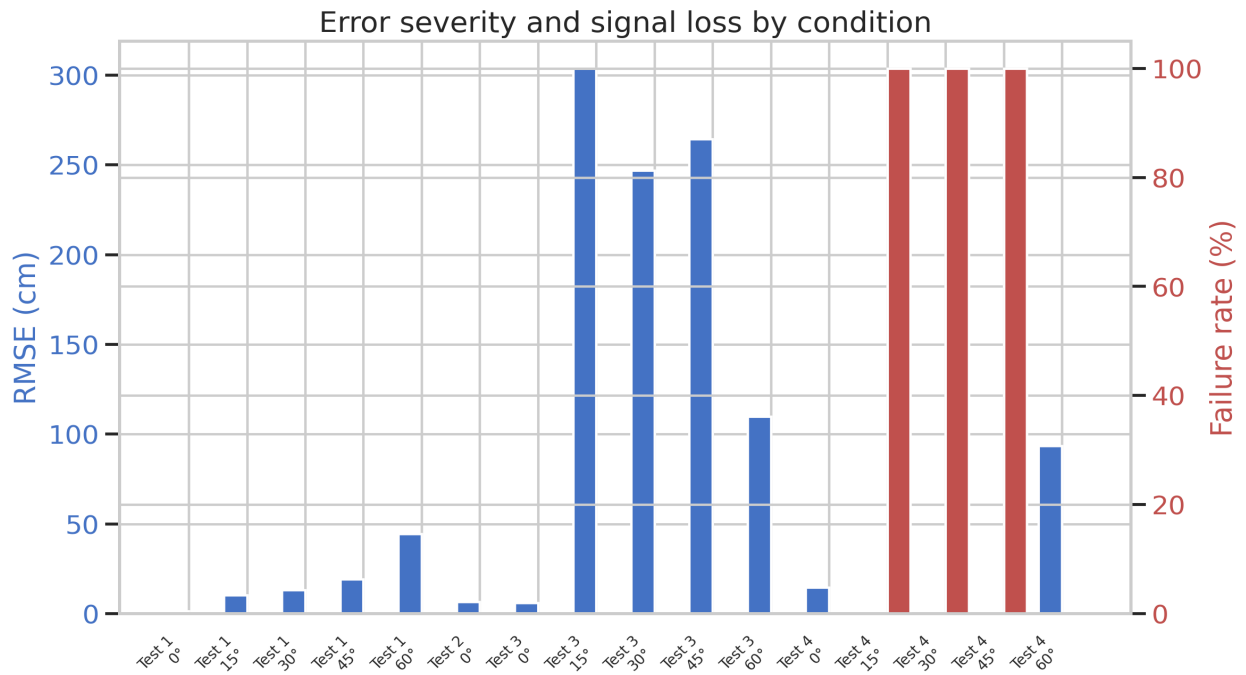
Test set	Angle	Actual (cm)	n valid	Failure %	Mean	Std	Min	Max	Bias	MAE	MAPE %	RMSE	Classification
Test 1	15°	42.50	31	0.00	52.90	0.83	51.00	55.00	10.40	10.40	24.48	10.44	Stable biased return
Test 1	30°	42.50	31	0.00	55.65	1.25	54.00	58.00	13.15	13.15	30.93	13.20	Severe biased return
Test 1	45°	42.50	31	0.00	61.48	2.92	54.00	68.00	18.98	18.98	44.67	19.20	Severe biased return
Test 1	60°	42.50	31	0.00	86.87	1.31	85.00	90.00	44.37	44.37	104.40	44.39	Catastrophic false ranging
Test 2	0°	30.10	30	0.00	36.63	1.47	34.00	38.00	6.53	6.53	21.71	6.69	Stable biased return
Test 3	0°	42.50	31	0.00	48.45	0.68	47.00	49.00	5.95	5.95	14.00	5.99	Stable biased return
Test 3	15°	42.50	31	0.00	346.29	1.79	343.00	350.00	303.79	303.79	714.80	303.80	Catastrophic false ranging
Test 3	30°	42.50	31	0.00	289.55	2.46	285.00	294.00	247.05	247.05	581.29	247.06	Catastrophic false ranging
Test 3	45°	42.50	31	0.00	307.06	0.85	306.00	309.00	264.56	264.56	622.50	264.57	Catastrophic false ranging
Test 3	60°	42.50	31	0.00	152.35	1.28	150.00	154.00	109.85	109.85	258.48	109.86	Catastrophic false ranging
Test 4	0°	42.50	31	0.00	56.90	2.37	52.00	60.00	14.40	14.40	33.89	14.59	Severe biased return
Test 4	15°	42.50	0	100.00	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Signal loss / no usable return
Test 4	30°	42.50	0	100.00	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Signal loss / no usable return
Test 4	45°	42.50	0	100.00	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Signal loss / no usable return
Test 4	60°	42.50	31	0.00	134.58	16.40	97.00	144.00	92.08	92.08	216.66	93.48	Catastrophic false ranging

Figure 1. Angle-dependent behaviour across tests



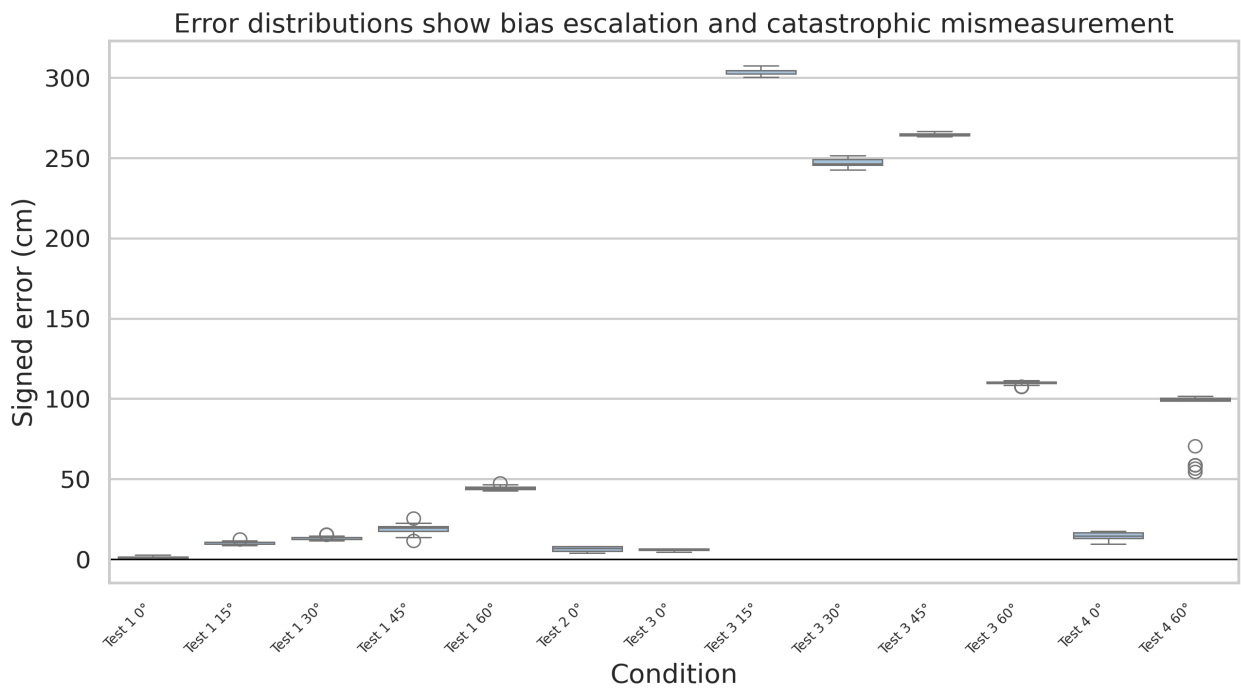
Mean measured distance rises modestly under transparent/reflective conditions but escalates dramatically for mirror-like conditions, especially at oblique angles.

Figure 2. Error severity and signal loss by condition



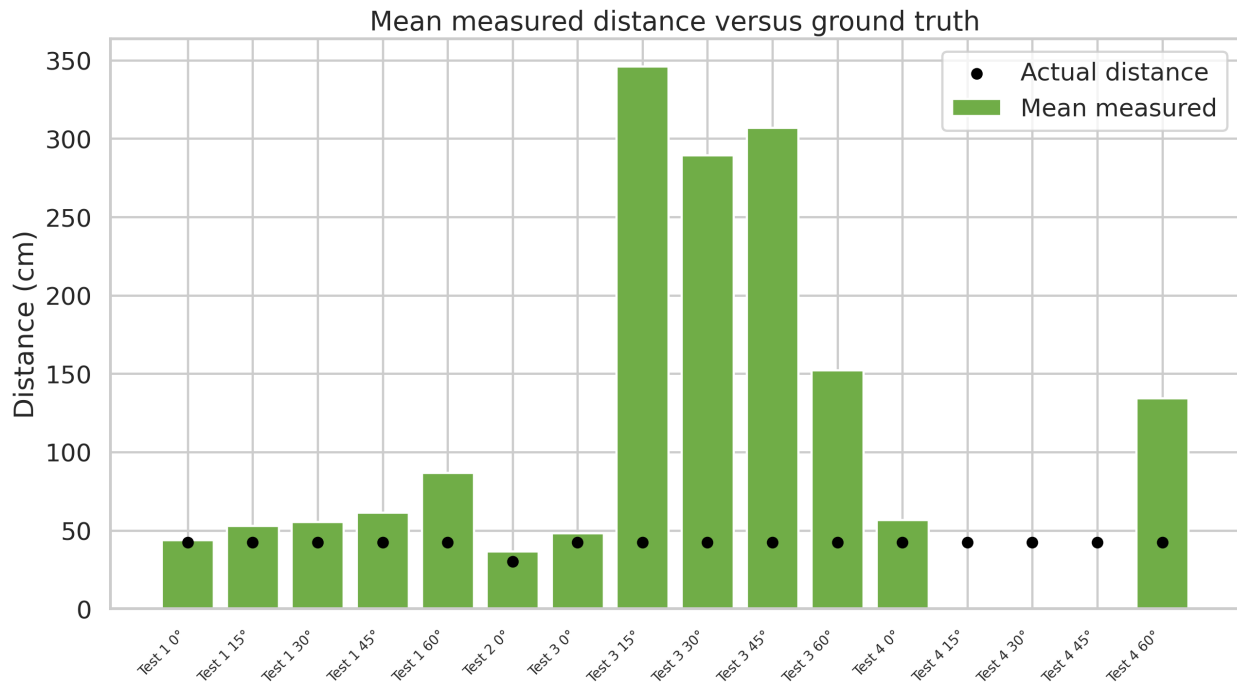
RMSE remains moderate for the transparent condition, becomes large in high-angle reflective conditions, and is extreme or undefined in mirror-loss regimes.

Figure 3. Distribution of signed errors



The boxplots distinguish stable positive bias from catastrophic over-ranging, highlighting that the principal issue is not only variance but the mean displacement of the return.

Figure 4. Mean measured distance versus ground truth



Several conditions generate readings far beyond the true separation, demonstrating the plausibility of severe state corruption in downstream navigation logic.

Results for Test 1

Test Set 1 shows a progressive angle effect. At 0°, the mean return is close to the actual 42.5 cm, indicating that the sensor can produce near-correct readings on this target when the beam is approximately normal to the surface. As incidence angle increases, the mean and RMSE increase monotonically, reaching a pronounced overestimate at 60°. This is consistent with an angle-sensitive reflective failure rather than random noise. The implication is that reflective or foil-like materials can remain deceptively benign at low angles while becoming operationally hazardous at higher angles.

Angle	Actual (cm)	n valid	Failure %	Mean	Std	Variance	Min	Max	Bias	MAE	MPE %	MAPE %	RMSE
0°	42.50	31	0.00	43.90	0.70	0.49	43.00	45.00	1.40	1.40	3.30	3.30	1.56
15°	42.50	31	0.00	52.90	0.83	0.69	51.00	55.00	10.40	10.40	24.48	24.48	10.44
30°	42.50	31	0.00	55.65	1.25	1.57	54.00	58.00	13.15	13.15	30.93	30.93	13.20
45°	42.50	31	0.00	61.48	2.92	8.52	54.00	68.00	18.98	18.98	44.67	44.67	19.20
60°	42.50	31	0.00	86.87	1.31	1.72	85.00	90.00	44.37	44.37	104.40	104.40	44.39

Results for Test 2

Test Set 2 demonstrates a transparent-surface vulnerability of a different kind. The readings are tightly clustered, indicating that the sensor is not simply failing randomly; rather, it is returning a repeatable but biased distance. This is security-relevant because stable false data may pass sanity checks that reject only high-variance anomalies. In a controller or estimator, a biased yet internally consistent return can persist long enough to influence altitude-hold or landing decisions.

Angle	Actual (cm)	n valid	Failure %	Mean	Std	Variance	Min	Max	Bias	MAE	MPE %	MAPE %	RMSE
0°	30.10	30	0.00	36.63	1.47	2.17	34.00	38.00	6.53	6.53	21.71	21.71	6.69

Results for Test 3

Test Set 3 is the clearest example of catastrophic false ranging. At 15°, 30°, and 45°, the mean readings are several times larger than the actual distance, with relatively small within-condition standard deviations. This means the sensor is consistently wrong, not merely unstable. Such conditions are dangerous because filtering based only on dispersion would not reliably reject them. The 60° condition remains catastrophically biased even though it is less extreme than 15°-45°.

Angle	Actual (cm)	n valid	Failure %	Mean	Std	Variance	Min	Max	Bias	MAE	MPE %	MAPE %	RMSE
0°	42.50	31	0.00	48.45	0.68	0.46	47.00	49.00	5.95	5.95	14.00	14.00	5.99

Angle	Actual (cm)	n valid	Failure %	Mean	Std	Variance	Min	Max	Bias	MAE	MPE %	MAPE %	RMSE
15°	42.50	31	0.00	346.29	1.79	3.21	343.00	350.00	303.79	303.79	714.80	714.80	303.80
30°	42.50	31	0.00	289.55	2.46	6.06	285.00	294.00	247.05	247.05	581.29	581.29	247.06
45°	42.50	31	0.00	307.06	0.85	0.73	306.00	309.00	264.56	264.56	622.50	622.50	264.57
60°	42.50	31	0.00	152.35	1.28	1.64	150.00	154.00	109.85	109.85	258.48	258.48	109.86

Results for Test 4

Test Set 4 extends the mirror result by introducing magnification. The reported behaviour bifurcates: at mid-angles, the image annotation indicates complete signal-quality loss, while at 60° the sensor returns large but fluctuating overestimates. This suggests that optical manipulation can drive the sensor into both denial-of-service-like failure and false-ranging behaviour. From a systems perspective, both outcomes are problematic: one removes a measurement channel, while the other supplies a misleading one.

Angle	Actual (cm)	n valid	Failure %	Mean	Std	Variance	Min	Max	Bias	MAE	MPE %	MAPE %	RMSE
0°	42.50	31	0.00	56.90	2.37	5.62	52.00	60.00	14.40	14.40	33.89	33.89	14.59
15°	42.50	0	100.00	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
30°	42.50	0	100.00	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
45°	42.50	0	100.00	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
60°	42.50	31	0.00	134.58	16.40	268.98	97.00	144.00	92.08	92.08	216.66	216.66	93.48

Cross-Test Comparison Analysis

Across the four sets, the most important cross-test distinction is that **different materials induce different classes of measurement failure**. Transparent material mainly produced a repeatable positive bias. Reflective/foil material produced a mild-to-severe positive bias that worsened with angle. Mirror-like material produced catastrophic over-ranging, with mean returns many times larger than the true distance. Mirror plus magnification additionally introduced a signal-loss region, showing that optical manipulation can move the sensor between false-data and no-data modes. The low-angle versus high-angle contrast is equally important. At or near 0°, several conditions remain close enough to truth that a designer might consider the sensor acceptable. However, the oblique-angle data show that this confidence does not generalise. In security terms, this means attack feasibility depends not only on material choice but also on geometry. A target or environmental object does not need to defeat the sensor in every pose; it only needs to induce harmful errors in plausible poses encountered during flight. Finally, the results separate **normal-return bias** from **catastrophic mismeasurement**. Test Set 2 would likely challenge systems that trust internally consistent data, whereas Test Sets 3 and 4 challenge systems that lack robust outlier rejection, signal-quality gating, or fallback logic. Both failure types are operationally significant, but they demand different defensive responses.

Failure mode	Observed in	Empirical signature	Likely impact on UAV logic
Stable biased return	Transparent surface; low-angle reflective	Low variance but persistent positive offset	Gradual altitude bias, premature landing flare, incorrect terrain estimate
Angle-driven degradation	Reflective/aluminium foil set	Error magnitude increases with incidence angle	Positional instability, intermittent navigation faults near glossy/metallic surfaces
Catastrophic false ranging	Mirror set; 60° magnified	Returns many times the actual distance	Obstacle proximity perceived much farther away than reality; estimator corruption
Signal loss / no return	Magnified mirror, 15°-45°	No usable signal-quality	Failure activation, stale data reuse, or estimator mode switching if not handled explicitly

Security Implications

In a GPS-denied UAV, a downward or forward rangefinder may influence altitude hold, terrain-relative motion, take-off and landing behaviour, collision prevention, or estimator updates. The present results show that apparently simple environmental materials can create states that are security-relevant even in the absence of active electronic spoofing. A transparent surface can cause the vehicle to believe it is farther from the surface than it really is. In a landing or low-altitude hold scenario, this could delay descent corrections or cause the vehicle to settle lower than intended before the estimator recognises the discrepancy. Reflective and foil-like materials introduce geometric sensitivity, meaning performance can deteriorate as the UAV attitude or the scene geometry changes. This raises the possibility of intermittent altitude excursions or inconsistent avoidance behaviour near polished floors, glossy structures, or metallic objects. The mirror conditions are more severe. A range estimate inflated from roughly 42.5 cm to 150-346 cm could cause the navigation stack to infer substantial free space where none exists. In a downward-looking configuration, the vehicle may believe it is much higher above ground than reality, potentially altering thrust or descent logic. In a forward-looking obstacle-avoidance role, false long-range returns can suppress avoidance responses. When magnification produces signal loss, the control problem shifts again: if the stack retains stale measurements, over-trusts the last good value, or falls back improperly, mission safety can still be compromised. For the dissertation's security framing, these observations support a threat model in which optical material manipulation functions as a low-cost attack primitive. The attack need not inject packets or compromise firmware; it can instead exploit the physics of return-path generation and the assumptions embedded in the perception pipeline.

Recommended Countermeasures

The countermeasures below are proposed in direct response to the observed failure modes. They are organised not merely as generic best practice, but as a mapping from empirical vulnerability to defensive function.

Countermeasure	Primary failure modes addressed	Rationale based on present results	Limitations / evaluation priority
Sensor fusion (LiDAR, stereo, IMU, etc.)	Stable bias, catastrophic jumps	Redundant modalities contradict implausible LiDAR estimates	Requires integrated sensor fusion logic; high priority for SITL evaluation
Temporal consistency checks	Catastrophic jumps, unstable high-angles	Consistent implausible trends can be rejected	Avoids false positives if drift is slow or constant
Angle-aware filtering	Angle-driven degradation	Error magnitude increases with incidence angle; geometry-aware models	Needs sensor or mounting geometry and attitude-aware implementation
Material-aware anomaly detection	Transparent, reflective, mirror, magnified	Stable bias vs extreme range inflation	Requires labelled data or learned models
Confidence / signal-quality gating	Signal loss and marginal returns	No-data states must be treated as missing information	Only if signal quality is exposed and logged consistently
Redundant sampling + robust estimators	Outliers (median filter, RANSAC, etc.)	Multiple samples can suppress sporadic false returns	Extensive testing on false ranging such as the mirror 15°-45° regime
Operational constraints	Highly reflective, polished, water-like	Known problematic materials	Reduces exposure on its own
Fallback navigation logic	Signal loss, degradation	Should hover, climb conservatively, or switch estimator	Needs explicit state transitions in code
Adversarial test expansion	Unknown materials	Broader testing across wet, IR-absorbing, glossy, curved, and transparent surfaces	Increases dissertation value also increases

Countermeasure Analysis

Based on the current data, no single countermeasure is sufficient. Robust filtering is useful against sporadic outliers, but the mirror results show that repeatable false ranges can be internally consistent and therefore survive purely statistical cleaning. Conversely, signal-quality gating can handle the Test 4 loss regime but does not solve Test 2's stable transparent-surface bias. The strongest near-term dissertation contribution would therefore come from combining three layers: (1) confidence gating to reject no-return states, (2) temporal and rate-of-change validation to reject physically implausible transitions, and (3) sensor fusion or estimator cross-checking so that a single optical channel cannot dominate altitude or obstacle decisions in isolation. In experimental terms, these three layers are also attractive because they are implementable and testable within PX4 SITL or a bench-top replay pipeline.

Conclusions

The uploaded test sets demonstrate that the Garmin LiDAR-Lite v4 can be manipulated by passive optical conditions into materially different error regimes: transparent surfaces produce stable overestimation, reflective/foil conditions produce angle-dependent degradation, mirror-like conditions produce catastrophic false ranging, and magnified mirror conditions can produce both severe overestimation and complete signal loss. For secure LiDAR-based UAV navigation in GPS-denied environments, this means the problem is not simply one of "noise" but of adversarially relevant failure semantics. A navigation stack that assumes a valid scalar range is inherently trustworthy is vulnerable to environmental deception. The practical recommendation is therefore to treat LiDAR as a conditional, confidence-managed sensor whose data must be cross-validated in time, geometry, and modality. From a dissertation standpoint, the strongest next step is to turn these bench observations into system-level evidence through SITL/HITL evaluation of selected countermeasures.

References

- Garmin. *LiDAR-Lite v4 LED Instructions*. Garmin Ltd. Available from Garmin product documentation.
- PX4 Autopilot Documentation. *Rangefinders and EKF2 tuning/configuration*. PX4 Developer Guide.
- Salguero-Luna, S. A., Ramírez-Gutiérrez, K. A., & Martínez-Cruz, A. (2025). *A State-of-the-Art Review on Attacks and Defense Mechanisms for LiDAR on Autonomous Vehicles*. IEEE Transactions on Intelligent Transportation Systems, 26(1), 22-42. <https://doi.org/10.1109/TITS.2024.3488432>
- Kobayashi, R., Nomoto, K., Tanaka, Y., Tsuruoka, G., & Mori, T. (2025). *Invisible but Detected: Physical Adversarial Shadow Attack and Defense on LiDAR Object Detection*. Proceedings of the 34th USENIX Security Symposium.
- Uploaded planning document: *Recommendations for Additional Work: Gap Analysis: Current Dissertation vs. 75%+ Model Dissertations*.
- Uploaded planning document: *Next Steps After Surface Testing — Action Plan*.

Appendix A. Detailed Measurement Tables

Appendix A1. Test 1 raw measurements

Trial	0°	15°	30°	45°	60°
1	43	55	55	60	87
2	45	53	56	62	85
3	45	52	58	63	87
4	43	53	54	65	87
5	44	53	55	60	87
6	44	52	56	64	89
7	45	53	57	61	86
8	43	52	54	60	89
9	44	53	56	58	87
10	43	54	56	61	86
11	45	52	56	56	89
12	44	54	55	61	86
13	44	53	58	63	85
14	44	52	56	65	86
15	44	53	54	63	88
16	43	52	56	64	86
17	43	53	58	59	86
18	44	51	54	61	86
19	44	53	57	62	87
20	44	54	56	63	86
21	44	53	55	63	85
22	44	53	57	63	86
23	43	53	55	62	86
24	45	54	55	58	86
25	44	53	54	54	86
26	44	54	57	59	87
27	43	52	55	61	89
28	44	52	54	58	88
29	44	53	56	64	88
30	45	53	54	65	87
31	43	53	56	68	90

Appendix A2. Test 2 raw measurements

Trial	Measured distance (cm)
1	37
2	37
3	38
4	35
5	35

Trial	Measured distance (cm)
6	38
7	34
8	34
9	35
10	34
11	38
12	38
13	37
14	35
15	37
16	38
17	38
18	37
19	37
20	38
21	37
22	35
23	35
24	35
25	38
26	38
27	38
28	38
29	38
30	37

Appendix A3. Test 3 raw measurements

Trial	0°	15°	30°	45°	60°
1	47	344	286	307	153
2	49	345	294	306	151
3	48	344	290	307	153
4	47	345	287	307	153
5	48	345	293	307	154
6	49	343	290	307	152
7	49	344	293	306	154
8	49	344	289	306	152
9	49	345	293	306	153
10	48	344	291	307	152
11	49	345	289	306	152
12	49	345	286	306	153
13	49	346	287	307	150

Trial	0°	15°	30°	45°	60°
14	47	346	290	306	150
15	49	347	292	306	150
16	48	347	288	307	152
17	49	347	289	307	154
18	49	346	292	309	152
19	49	347	288	308	153
20	49	347	287	307	152
21	49	347	288	307	151
22	49	347	293	308	153
23	48	347	288	308	151
24	48	348	290	307	150
25	48	349	293	307	154
26	48	348	291	307	153
27	48	349	289	309	154
28	48	348	285	307	153
29	49	349	290	308	154
30	49	347	288	308	153
31	48	350	287	308	152

Appendix A4. Test 4 raw measurements and signal-loss notes

Trial	0°	15°	30°	45°	60°
1	56	signal quality lost	signal quality lost	signal quality lost	143
2	53	signal quality lost	signal quality lost	signal quality lost	143
3	52	signal quality lost	signal quality lost	signal quality lost	142
4	60	signal quality lost	signal quality lost	signal quality lost	101
5	58	signal quality lost	signal quality lost	signal quality lost	99
6	55	signal quality lost	signal quality lost	signal quality lost	143
7	55	signal quality lost	signal quality lost	signal quality lost	143
8	56	signal quality lost	signal quality lost	signal quality lost	142
9	57	signal quality lost	signal quality lost	signal quality lost	143
10	57	signal quality lost	signal quality lost	signal quality lost	113
11	59	signal quality lost	signal quality lost	signal quality lost	101
12	52	signal quality lost	signal quality lost	signal quality lost	142
13	52	signal quality lost	signal quality lost	signal quality lost	97
14	59	signal quality lost	signal quality lost	signal quality lost	101
15	59	signal quality lost	signal quality lost	signal quality lost	143
16	59	signal quality lost	signal quality lost	signal quality lost	142
17	55	signal quality lost	signal quality lost	signal quality lost	143
18	59	signal quality lost	signal quality lost	signal quality lost	141
19	59	signal quality lost	signal quality lost	signal quality lost	142
20	59	signal quality lost	signal quality lost	signal quality lost	144

Trial	0°	15°	30°	45°	60°
21	59	signal quality lost	signal quality lost	signal quality lost	143
22	57	signal quality lost	signal quality lost	signal quality lost	141
23	58	signal quality lost	signal quality lost	signal quality lost	142
24	56	signal quality lost	signal quality lost	signal quality lost	142
25	56	signal quality lost	signal quality lost	signal quality lost	142
26	57	signal quality lost	signal quality lost	signal quality lost	142
27	55	signal quality lost	signal quality lost	signal quality lost	143
28	60	signal quality lost	signal quality lost	signal quality lost	143
29	58	signal quality lost	signal quality lost	signal quality lost	143
30	58	signal quality lost	signal quality lost	signal quality lost	141
31	59	signal quality lost	signal quality lost	signal quality lost	142

Image annotation for Test 4: "15 to 55 degrees signal quality is lost"; "from angle of 60+ to 85 average distance tends to value of 82 cm." The uploaded table includes a 60° column with the measurements analysed here.